Computer science lesson plan: can your classmates escape a world you invented?

Draft for Tomer's review

Not tried with children. Timings are targets: nothing has been timed with a class, and the Lovable build wait for eight pairs at once is not measured.

A US-style computer science lesson plan built around the CSTA K-12 Computer Science Standards. The activity sequence is Tomer's flow: a paper idea card first, then Lovable inside imagi Edu. Minute-by-minute words to say are in lesson-plan-tom-flow.md.

Grade	7 (CSTA level 2, grades 6-8)
Time	45 minutes, plus an optional 15-minute extension
Class	16 learners in 8 pairs, by learner number. Partner A is the odd number, Partner B the even one
Devices	8 shared laptops with Chrome and Wi-Fi; 1 projector laptop
Tools	Paper cards; Lovable inside imagi Edu
Demo game	Alien Snack Shop, on the projector: https://imagi-mockup.vercel.app/alien-snack-shop-v1.html. A hand-built example of what the starting prompt asks Lovable to make, not a Lovable build

Standards addressed

Code	Standard	Where in the lesson
CSTA 2-AP-13	Decompose problems and subproblems into parts to facilitate the design, implementation, and review of programs	Idea card: world, clue, code, ending, colours and mood (5-8)
CSTA 2-AP-15	Seek and incorporate feedback from team members and users to refine a solution that meets user needs	Playtest and choosing a change from what the player did (17-31)
CSTA 2-AP-17	Systematically test and refine programs using a range of test cases	Prediction, Version 1 test, retest of the change and a kept rule (8-17, 31-35)
CSTA 2-AP-18	Distribute tasks and maintain a project timeline when collaboratively developing computational artifacts	Partner A types Version 1, Partner B types Version 2; timed blocks
CSTA 2-AP-19	Document programs in order to make them easier to follow, test, and debug	Change / Because / Keep / Check; the maker note
ISTE Students 1.4	Innovative Designer: students use a variety of technologies within a design process to identify and solve problems by creating new, useful or imaginative solutions	The whole build, test and improve cycle
ISTE Students 1.5	Computational Thinker: students develop and employ strategies for understanding and solving problems in ways that leverage the power of technological methods to develop and test solutions	Instructing the AI and testing what it built
CCSS ELA SL.7.1	Engage effectively in a range of collaborative discussions (one-on-one, in groups, and teacher-led) with diverse partners on grade 7 topics, texts, and issues, building on others' ideas and expressing their own clearly	Choosing a change with a partner; explaining a choice to the player

CSTA codes are from the K-12 Computer Science Standards (2017), level 2. Learners don't write code: Lovable writes it from their plain-language instructions. The alignment is to designing, instructing, testing and refining, not to programming syntax.

2-IC-20 and 2-IC-21 were considered and left out: the lesson doesn't compare tradeoffs or discuss bias. Check the wording against the current version before formal use.

Learning objectives

Students will be able to:

1. Break a game idea into parts: world, clue, code, ending, colours and mood (2-AP-13).
2. Predict what their game will do before they test it (2-AP-17).
3. Watch a real player and write down what the player did, not what they should have done (2-AP-15).
4. Choose one change for a reason and explain it to a partner (2-AP-15, SL.7.1).
5. Give the AI a clear instruction that says what to change and what to keep (2-AP-19).
6. Test the change and one rule that should still work, against their prediction (2-AP-17).

Vocabulary

Word	Meaning for learners
Clue	Something in the room that helps the player find the code
Code	The secret answer that opens the exit
Hint	Help that points to where to look, without giving the code
Player	The person who tries to escape. Not the person who made it
Prediction	What you think will happen, written before you test
Test	Trying the game yourself to see what really happens
Version	The game at one point in time. Version 2 is the game after your change
Instruction	What you tell the AI to do, in your own words
Keep	The parts the AI must not change
"Done"	What the AI says when it finishes. It is a claim until you test it

Materials

Item	How many	File
Idea card	8, one per pair	kit/student/idea-card.md; PDF: kit/student/pdf/idea-card.pdf
Playtest card	8, one per pair	kit/student/playtest-card.md; PDF: kit/student/pdf/playtest-card.pdf
Starting prompt in a shared document	1	lovable/starting-prompt.md; [link from Liba]
Student guide slides	1, on the projector	kit/student/guide.md
Help card	8	kit/student/help-card.md
Checks A and B	16 each	kit/student/checks.md
Maker note	16	kit/student/maker-note.md
Take-home card	16	take-home-card.md
Paper backup	2 sets	paper-backup.md
Scoring guide and observation sheet	1	scoring-guide.md

Item	How many	File
Pencils	16	

Preparation takes under 20 minutes. The steps and their status are in lesson-plan-tom-flow.md, "Setup before class".

Warm-up (0-5 minutes)

Min	Teacher	Students
0-2	Open Alien Snack Shop on the projector. Say: "Can your classmates escape a world you invented? Today you'll build one, test it on a classmate and make it better." Invite one volunteer by learner number. If they're stuck: "Want a hint, or one more try?"	The volunteer plays: a wrong code, Hint, the receipt and the lock pictures, the right code, the snack party. The class calls out guesses
2-5	Hand out the starting check: odd numbers A, even numbers B. Say: "On your own. Write what you think. There's no trick."	Starting check alone, up to 3 minutes. No partner, no AI

Main activities (5-35 minutes)

Min	Activity	Keyboard	Teacher	Students	Standard
5-8	Plan on paper	Nobody	"Your idea first. The AI comes after. Each of you writes at least one line." Check each idea is suitable for school	Fill the idea card: world, clue, code, ending, colours and mood	2-AP-13
8-10 (target, untimed. reference/assignment.md asks for a first playable game by minute 10. With the prompt sent at 8-10 and up to 3 minutes of build, it may arrive at 11-13. Loom 1 times one build; it still needs timing with children)	Instruct the AI	Partner A	"Copy the prompt, fill the five blanks from your card, and send it. While it builds, each of you writes one prediction."	In imagi Edu, open Lovable; new project "Escape - pair N"; paste the starting prompt; fill the five blanks; send. Both write: "When I type a wrong code, I think the game will ___."	2-AP-17, ISTE 1.5
10-17	Test Version 1 and make it theirs	Partner A	"Was your prediction right? What would make it feel more like your world?"	Play: things, a wrong code, an empty code, Hint, the right code, Restart. Send one change in their own words. Play it again	2-AP-17
17-23	User test	The player	Pairs 1 and 2 swap, 3 and 4, and so on. "Players, play. Creators, no helping and no pointing." At 20: "Swap."	One learner plays the neighbours' game while a creator watches and writes: "The player clicked ___ first. They got stuck at ___."	2-AP-15

Min	Activity	Keyboard	Teacher	Students	Standard
23-31	Refine from feedback	Partner B	"What did your player do? Which change helps them? Why that one?"	Each child proposes a change. The pair picks one and writes Change / Because / Keep / Check. Partner B sends it. Both predict what a new player will do	2-AP-15, 2-AP-19, SL.7.1
31-35	Retest	Partner B	"Test the change, then one thing that should still work." "Lovable says it's done. What happened when you tried it?"	Test the change, then a kept rule. Write the result next to the prediction. If there's time, a new player tries it	2-AP-17, ISTE 1.4

No playable game by minute 12: the pair moves to the paper backup. The help steps for every stuck moment are in lesson-plan-tom-flow.md, "When someone is stuck".

Closure (35-45 minutes)

Min	Teacher	Students
35-38	Hand out the other check. "Same as before: on your own."	The other check, alone
38-45	"Close your Lovable tab. Open your project again from imagi and check it's still there. What would you add next?" If a minute is left, ask two pairs: "What did you change that made it work better?"	Reopen the project from imagi (not tested on the student route). Fill the maker note and the take-home card: the game, one choice we made, the next idea

Assessment

Formative, during class

Evidence	When	Look for
Idea card	5-8	At least one line from each partner, in their own words
Prediction line	8-10, 23-31	Written before the test
Playtest line	17-23	What the player did, not what they should have done
Change / Because / Keep / Check	23-31	The change comes from what the player did; says what must stay
Retest result	31-35	The change and a kept rule, compared with the prediction
"What happened when you tried it?"	Any time	They describe what they saw, not what Lovable said

Record per pair on the per-pair sheet in scoring-guide.md: first playable minute, paper backup used, adult rescues, build waits, the change chosen and why, and whether a player asked to make their own game.

Checks A and B

Each child does one check at the start (2-5) and the other at the end (35-38), alone, up to 3 minutes each. Odd learner numbers do A then B; even numbers do B then A. Record any reading support. Keep the first answer before any coaching.

Check	0	1	2
Reason (Q1)	No relevant reason	Spots the failure	Uses the result and names an untested rule
Instruction (Q2)	Vague or breaks a rule	Specifies the change	Specifies the change and what to keep
Test (Q3)	Only "test it"	An action and expected result	Checks the fix and a kept rule, with expected results

Score each child alone with scoring-guide.md. A pair's game is not an individual score. Missing answers stay missing, not zero. These are classroom checks, not validated measures. The maker note shows each child's own account: "The player did ___. We changed ___. The next test showed ___."

Differentiation

Learners	Support
Need reading support	Read check questions aloud on request and note it on the sheet. The help card has one action per line. Children fill five blanks, not a whole prompt. A child may say or sketch an idea; the partner or teacher writes it in the child's words
English learners	Put the vocabulary on the board with a small drawing for each word. Sentence frames on the playtest card ("The player clicked ___ first.") give the structure. A pair may plan in their home language, then put the five blanks into English with the teacher, keeping the child's words. Not tested: how Lovable handles blanks in other languages
Early finishers	A second player who hasn't seen the game tries it, and the pair writes what happened. Or one step up the challenge ladder (below). Or write the invitation: "Can you escape my ___?"
Learners who struggle	Ask one question at a time and wait. Keep to one change. The paper backup lets a pair choose, play, change and retest without a laptop. The teacher may scribe, but never fills a blank with an idea

Extensions

The optional 15 minutes are for an upgrade or a second test, never for finishing the core. Next challenges, one at a time:

1. A second clue that the player needs before the code works.
2. A new rule: click things in the right order.
3. Two players taking turns on one laptop. No second device and no link.
4. Freestyle: the pair's own kind of game, still one screen.

Cross-curricular links

Area	Where it shows up
English (speaking and listening)	Explaining a choice to a partner and to the player (SL.7.1)
Values: respect and fair play	Taking turns at the keyboard; not helping or pointing while a classmate plays
Mathematics	Number codes, order, counting clues (only if the learners choose a number code)
Digital safety	Learner numbers, never names; no personal details in AI; sharing only through an adult

These are links to other areas, not extra learning claims.